

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements(FR)** define specific behaviors, Functions or features or the system(What system should do).
- **Non-Functional Requirements(NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors(How the system should be).

Functional Requirements:

Category	Description	Priority
Authentication	Users can securely register, log in, and access the application using secure authentication.	High
Authorization	Only authenticated users can upload audio, access evaluation reports, and view their learning history.	High
Audio Input	The system allows users to upload or record voice explanations in supported audio formats.	High
Speech-to-Text	The system converts uploaded audio into text using the Whisper Speech-to-Text model.	High
Semantic Analysis	The system compares the transcript with the reference concept using Sentence-BERT to	High

	calculate semantic similarity.	
Audio Feature Analysis	The system analyzes pause ratio, RMS energy, filler words, speech rate, and other audio quality metrics using Librosa.	High
Evaluation & Scoring	The system generates a concept understanding score and provides qualitative feedback based on semantic and audio analysis.	High
Reporting		Medium

Non-Functional Requirements

Category	Requirement	Target
Performance	Audio processing and evaluation should complete quickly.	Response time below 5 seconds for normal audio files
Scalability	The system should support multiple concurrent users without performance degradation.	Supports multiple concurrent users
Security	User data, uploaded audio, and reports must be securely stored and protected.	Secure authentication & encrypted data
Reliability	The application should provide accurate evaluations with minimal processing failures.	99.9% uptime
Usability	The interface should be simple, responsive, and easy to use on desktop devices.	User-friendly interface
Maintainability	The application should follow a modular architecture for future updates and enhancements.	Easy maintenance & upgrades
Compatibility	Support common browsers and standard	Cross-browser compatibility

	audio file formats (WAV, MP3, M4A).	
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